PROCESSORS Undergoing Changes To SAVE POWER

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Power versus performance is an ongoing battle, with the best involving a compromise between the two. High-performance system(s)-on-chip (SoC) designers often struggle to achieve the best possible performance-power balance. Customers' user experience is an important aspect to consider as it determines performance improvements. However, this calls for increasing the battery size as most devices are battery-powered. Memory tracker technology from Performance-IP, for example, enables products to operate at higher efficiency, while reducing power consumption.

Power reduction techniques for processors

A typical SoC design consists of processors, memory clients, interconnects and memory systems. Processors are one of the major power consumers in a device, so switching off the CPU when idle helps. This includes turning off the processor when all processes are blocked, when processes appear to be busy waiting and extending real-time process sleep periods. Studies have shown these to increase battery lifetime by approximately 20 per cent in Apple systems.

These tricks are often used by processor designers as well. Dynamic power consumption in a processor arises from circuit activity. As performance, that is, speed and frequency, of the IC increases, the amount of dynamic power also increases. Dynamic power is data-dependent and closely tied to the number of transistors that change states. A hidden component of dynamic power is loss due to dynamic hazards. This must be taken care of while designing power-conscious processors.

Architectural changes to reduce power. Crusoe from Transmeta is a very long instruction word (VLIW) processor designed for low-power applications including mobile PCs and Internet devices. It allows microprocessors to emulate Intel X86 instruction set. Instead of the instruction set being implemented on the hardware, Crusoe runs a software abstraction layer known as code morphing software.

This has brought about a reduction in required power. Intel processors at 400MHz used to consume about 7.4 watts, or AMD clocked at 700MHz at the time consumed 34 watts. The 700MHz Crusoe uses one watt, which was a revelation when introduced.

LongRun in Crusoe monitors the precise performance level needed by an application and dynamically adjusts the processor's operating speed and voltage to match it. This increases the battery life by a significant amount.

Following this, there is SpeedStep

Fig. 1: Zen Core block diagram